| INCH-POUND |

MIL-M-38510/111
AMENDMENT 7
7 July 1992
SUPERSEDING
AMENDMENT 6
5 August 1985

MILITARY SPECIFICATION

MICROCIRCUITS, ANALOG SWITCH WITH DRIVER,
MONOLITHIC AND MULTI-CHIP SILICON

This amendment forms a part of Military Specification MIL-M-38510/111, dated 3 January 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 scope, line 2: Delete "Three" and substitute "Two".

* 1.2.3: Add case outline letter "X", F-2A (14-lead, .390" x .260" x .115"), flat package.

PAGE 2

1.4, line 5: Delete "Case operating" and "T_C" (as substituted by amendment 2) and substitute "Ambient" and "T_A", respectively.

1.5, maximum allowable power dissipation column: Delete "T_C" (3 places) (as substituted by amendment 2) and substitute "T_A".

PAGE 3

3.4, line 2: Delete "case" (as substituted by amendment 2) and substitute "ambient".

3.4.1, line 1: Delete "are guaranteed to" and substitute "shall".

3.5, electrical test requirements, delete (as substituted by amendment 3) and substitute the following:

"3.5 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

3.7, line 2: Delete "57" and substitute "94".

PAGE 4

TABLE I, $I_{S(OFF)}$ and $I_{D(OFF)}$ tests, min limits column: Delete "-10" (added in amendment 2) and substitute "-100" (6 places).

TABLE I, $I_{D(ON)} + I_{S(ON)}$ test, min limits column: Delete "-20" (added in amendment 2) and substitute "-200" (2 places).

TABLE I, temperature column: Delete "T_C" (12 places) (as substituted by amendment 2) and substitute "T_A".

PAGE 5

TABLE I, temperature column: Delete "T_C" (17 places) (as substituted by amendment 2) and substitute "T_A".

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PAGE 6

TABLE I, crosstalk between channels, V_{CT}, min limits column: Delete "50" and substitute "60".

TABLE I, temperature column: Delete "T_C" (7 places) (as substituted by amendment 2) and substitute "T_A".

PAGE 7

TABLE II, delete (as substituted by amendment 4) and substitute the following:

"TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9	1*,2,3,9
Group A test requirements (method 5005)	1,2,3, (4,7)***,9, 10,11,12	1,2,3, (4,7)***,9, 10,11,12
Group C end-point and group B, class S electrical parameters (method 5005)	1,2,3 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	Not applicable	(4,7)****
Group D end-point electrical parameters (method 5005)	1,2,3	1
Additional electrical subgroups for group D periodic inspections	(4,7)**	None

* PDA applied to subgroup 1 (see 4.3d).
** See 4.4.4b.
*** See 4.4.1d.
**** See 4.4.3c."

4.3, add new subparagraph f as follows:

"f. Burn-in circuit resistor tolerances for figures 3, 4, and 5 shall be ±10 percent."

4.3d, line 8, following "subgroup 1 after burn-in": Add "divided by the total number of devices submitted for burn-in".

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4.3e, following "2010": Add "and method 2017 multichip criteria".

4.4.1b, following "table II": Add "herein".

4.4.1, add subparagraphs c and d as follows:

"c. Subgroup 12 (device types 05, 06, 07, and 08) shall be added to group A inspection as specified in table III herein. The LTPD for subgroup 12 shall be 5 for all classes.

d. Subgroups 4 and 7 shall be performed for initial qualification only using a sample of 5 devices for each device type submitted to group A inspection, with no failures allowed. If not more than 1 failure is found in the first sample of 5, a second sample of 5 is permitted with no further failures allowed."

4.4.1c, line 3: Delete "5" (as substituted by amendment 4) and substitute "7".

4.4.2b, delete and substitute:

"b. Life test for class S shall be performed as specified in subgroup 5 of table IIa of method 5005 of MIL-STD-883 using circuit shown on figure 5. If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used for device types 05, 06, 07, and 08; life test duration shall be divided equally between forward and reverse bias."

PAGE 8

* Terminal connections: Delete "Cases A and D" and substitute "Cases A, D, and X".

PAGE 9

* Terminal connections: Delete "Cases A and D" and substitute "Cases A, D, and X".

PAGE 10

* Terminal connections: Delete "Cases A and D" and substitute "Cases A, D, and X".

PAGE 11

* Terminal connections: Delete "Cases A and D" and substitute "Cases A, D, and X".

PAGES 22, 23, AND 24

FIGURES 6, 7, and 8: Delete the "30 pF" value for the test circuit capacitors and substitute "100 pF".

PAGE 25

FIGURE 9, note 1b: Delete "(0.3 V to 0.7 V)" and substitute "(0.3 V to 2.7 V)".

PAGE 26

FIGURE 9, note 3: Delete "c, d, e, and f" for devices 01, 03, 05, and 07. Also, delete "c, d, e, and f" for devices 02, 04, 06, and 08.

FIGURE 9, load capacitor C_L: Delete "100 pF" and substitute "(95-200) pF".

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PAGE 27

FIGURE 10, on the illustrated circuit, above "CL2", next to "V_O", add "V_A"; above "CL1", next to "V_O", add "V_B".

FIGURE 10, on the second waveform from the top: Delete "A" (2 places) and substitute "V_A"; on the third waveform: Delete "B" (2 places) and substitute "V_B ".

PAGE 28

FIGURE 10, note 1, line 2, delete and substitute: "$C_{L1} = C_{L2}$ = (95-200) pF ±5%, to include test and fixture capacitance."

FIGURE 10, note 6, line 3: Delete "$\leq$ 5 ns" and substitute "$\geq$ 5 ns".

PAGE 29

TABLE III, test 11, adapter pin number 11 column: Delete "-10.0" (added in amendment 1) and substitute "7.5"; test 12, adapter pin number 12 column: Delete "-10.0" (added by amendment 1) and substitute "-7.5". Make the same changes on page 30, tests 35 and 36; and on page 31, tests 59 and 60.

TABLE III, tests 11 and 12, limits column: Delete "-200" and "200" (as substituted by amendment 4) and substitute "-2" and "2", respectively. Make this same change on page 33, tests 11 and 12; page 37, tests 21 through 24; page 43, tests 21 through 24; page 49, tests 11 and 12; page 53, tests 11 and 12; page 57, tests 21 through 24; and on page 63, tests 21 through 24.

PAGE 30

TABLE III, subgroup column: Delete "T_C" (as substituted by amendment 2) and substitute "T_A". Make this same change on pages 31 through 68.

TABLE III, min limits column, tests 27-34 and 51-54: Delete "-10" (added in amendment 2) and substitute "-100" (12 places). Make the same changes on page 31, tests 55-58; page 33, tests 27-29; page 34, tests 30-34 and 51-58; page 38, tests 41-56; page 39, tests 77-84; page 40, tests 85-92; page 44, tests 41-56; page 45, tests 77-84; page 46, tests 85-92; page 49, tests 25-28; page 50, tests 29-32 and 47-54; page 53, tests 25 and 26; page 54, tests 27-32 and 47-54; page 58, tests 41-56; page 59, tests 77-84; page 60, tests 85-92; page 64, tests 41-53; page 65, tests 54-56 and 77-79; page 66, tests 80-92.

TABLE III, min limits column, tests 35 and 36: Delete "-20" (added in amendment 2) and substitute "-200" (2 places). Make the same changes on page 31, tests 59 and 60; page 34, tests 35 and 36; page 35, tests 59 and 60; page 39, tests 57-60; page 40, tests 93-96; page 45, tests 57-60; page 46, tests 93-96; page 50, tests 33, 34, 55, and 56; page 54, tests 33 and 34; page 55, tests 55 and 56; page 59, tests 57-60; page 60, tests 93-96; page 65, tests 57-60; page 66, tests 93-96.

PAGE 31

TABLE III, test 77, adapter pin number 14 column: Add "2.0 V".

PAGE 33

TABLE III, test 11, adapter pin number 11 column: Add "-10 V"; test 12, adapter pin number 12 column: Add "-10 V". Make the same changes on page 34, tests 35 and 36; and on page 35, tests 59 and 60.

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PAGE 35

TABLE III, test 77, adapter pin number 14 column: Add "2.0 V".

PAGE 37

TABLE III, test 21, adapter pin number 10 column: Add "-7.5 V"; test 22, adapter pin number 11 column: Add "-7.5 V"; test 23, adapter pin number 12 column: Add "-7.5 V"; test 24, adapter pin number 13 column: Add "-7.5 V". Make the same changes on page 39, tests 57, 58, 59, and 60; and on page 40, tests 93, 94, 95, and 96.

PAGE 43

TABLE III, test 21, adapter pin number 10 column: Add "-10 V"; test 22, adapter pin number 11 column: Add "-10 V"; test 23, adapter pin number 12 column: Add "-10 V"; test 24, adapter pin number 13 column: Add "-10 V". Make the same changes on page 45, tests 57, 58, 59, and 60; and on page 46, tests 93, 94, 95, and 96.

PAGE 46

TABLE III, test 99 and 100, min limits column: Add "-1".

PAGE 49

TABLE III, test 11, adapter pin number 12 column: Add "-7.5 V"; test 12, adapter pin number 11 column: Add "-7.5 V". Make the same changes on page 50, tests 33 and 34; and tests 55, and 56.

PAGE 51

TABLE III, test 67, adapter pin number $3S_1$ column: Add "GND"; test 68, adapter pin number $4S_2$ column: Add "GND".

TABLE III, test 69 (calculation), min limits column: Delete "50" and substitute "60"; test 71 (calculation), min limits column: Delete "60" and substitute "50".

PAGE 52

TABLE III, test 85, measured pin no. column: Delete "11" and substitute "12".

PAGE 53

TABLE III, test 11, adapter pin number 12 column: Add "-10 V"; test 12, adapter pin number 11 column: Add "-10 V". Make the same changes on page 54, tests 33 and 34; and on page 55, tests 55 and 56.

PAGE 54

TABLE III, test 49, adapter pin number 3 column: Add "10 V". Adapter pin number 11 column: Add "-10 V".

TABLE III, test 50, adapter pin number 4 column: Add "10 V". Adapter pin number 12 column: Add "-10 V".

TABLE III, test 53, adapter pin number 3 column: Add "-10 V". Adapter pin number 11 column: Add "10 V".

TABLE III, test 54, adapter pin number 4 column: Add "-10 V". Adapter pin number 12 column: Add "10 V".

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PAGE 55

TABLE III, test 67, adapter pin number $3S_1$ column: Add "GND"; test 68, adapter pin number $4S_2$ column: Add "GND".

TABLE III, test 69 (calculation), min limits column: Delete "50" and substitute "60". Test 71 (calculation), min limits column: Delete "60" and substitute "50".

PAGE 56

TABLE III, test 85, measured pin no. column: Delete "11" and substitute "12".

PAGE 57

TABLE III, test 21, adapter pin number column 10: Add "-7.5 V"; test 22, adapter pin number 11 column: Add "-7.5 V"; test 23, adapter pin number 12 column: Add "-7.5 V"; test 24, adapter pin number 13 column: Add "-7.5 V". Make the same changes on page 59, test 57, 58, 59, and 60; and on page 60, tests 93, 94, 95, and 96.

PAGE 61

TABLE III, test 116 (calculation), max limits column: Delete "1.0".

PAGE 63

TABLE III, test 21, adapter pin number column 10: Add "-10 V"; test 22, adapter pin number 11 column: Add "-10 V"; test 23, adapter pin number 12 column: Add "-10 V"; test 24, adapter pin number 13 column: Add "-10 V". Make the same changes on page 65, test 57, 58, 59, and 60; and on page 66, tests 93, 94, 95, and 96.

PAGE 69

4.4.3a, following "table II": Add "herein".

4.4.3c, delete and substitute:

"c. A special subgroup shall be added to group C inspection for classes B and C devices only, and it shall consist of the group A subgroups 4 and 7 as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. A sample of 5 devices (of the device type to be inspected) shall be chosen and submitted to test with no failures allowed. If not more than 1 failure is found in the first sample of 5, a second sample of 5 is permitted with no further failures allowed. When more than one device type is qualified, the single device type selected shall be a different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfill the periodic inspection requirement."

Delete 4.4.4b and 4.4.4c (as substituted by amendments 2 and 1, respectively) and substitute 4.4.4b as follows:

"b. A special subgroup shall be added to group D inspection for class S devices only, and it shall consist of the group A subgroups 4 and 7 as specified in table III herein. This special subgroup shall be performed on each device type that is qualified from those listed in 1.2.1 herein. After initial qualification, the special subgroup shall be performed periodically on a single device type selected from those device types previously qualified. When more than one device type is qualified, the single device type selected shall be a different device type for each subsequent periodic inspection until all qualified device types have been inspected. The sequence of single device types shall be repeated to fulfill the periodic inspection requirement."

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TABLE IV, line 1 of the title: Delete "Groups C and D end-point electrical parameters" and substitute "Group C end-point electrical parameters".

TABLE IV, test no. column: Delete test no. column and all test numbers.

TABLE IV, test $I_{S(OFF)}$ and $I_{D(OFF)}$, device types 01, 03, 05, 07 min limits column: Delete "-1 mA" and substitute "-1 nA".

PAGE 70

6.3b, line 1: Delete "qualify" and substitute "quality".

PAGE 71

6.6, generic-industry type column: Delete "DG" (8 places).

Concluding material: Delete and substitute as printed in this amendment.

The margins of this amendment are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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